

PHARMACOLOGY

NCLEX STUDY GUIDE

Medication families • Adverse effects • Antidotes • Safety

1. Core Medication Safety

Principle	NCLEX application
Before giving	Verify patient, drug, dose, route, time, indication, allergies, interactions, relevant labs and ability to receive the medication.
After giving	Document administration and evaluate the intended response, adverse effects and need for follow-up.
High-alert drugs	Use independent double-checks according to policy for insulin, anticoagulants, opioids, chemotherapy, concentrated electrolytes and other high-alert agents.
Trough / peak	Trough is drawn immediately before the next dose; peak timing depends on route and medication. Follow the exact order.
Do not crush	Avoid crushing enteric-coated, sustained/extended-release, sublingual or hazardous medications.

PRIORITY When a serious adverse reaction is suspected: stop or hold the medication when appropriate, assess ABCs and vital signs, notify the provider/rapid response, and prepare the antidote or emergency treatment.

High-yield antidotes

Exposure	Antidote / immediate treatment
Opioids	Naloxone; support airway and ventilation.
Benzodiazepines	Flumazenil in selected cases; seizure risk limits routine use.
Acetaminophen	N-acetylcysteine.
Heparin	Protamine sulfate.
Warfarin	Vitamin K; serious bleeding may require clotting-factor replacement.
Digoxin	Digoxin immune Fab.
Magnesium toxicity	Calcium gluconate.
Cholinergic toxicity	Atropine plus pralidoxime for organophosphate poisoning.
Anticholinergic toxicity	Supportive care; physostigmine only in selected severe cases.
Iron	Deferoxamine.
Lead	Succimer or other chelation according to severity.

2. Respiratory Medications

Class / examples	Use	Key effects / teaching
SABA: albuterol, levalbuterol	Acute bronchospasm/rescue	Tremor, tachycardia and hypokalemia; assess response.
LABA: salmeterol, formoterol	Maintenance	Not a substitute for rescue therapy; in asthma use with inhaled corticosteroid.
Anticholinergic: ipratropium, tiotropium	COPD maintenance; adjunct in asthma	Dry mouth, urinary retention, blurred vision; avoid spraying into eyes.
Methylxanthine: theophylline	Selected chronic airway disease	Narrow range; nausea, insomnia, tachyarrhythmia or seizures suggest toxicity. Numerous interactions.
Inhaled corticosteroid	Long-term asthma control	Rinse mouth; monitor thrush/hoarseness. Not rescue therapy.
Systemic corticosteroid	Exacerbation/inflammation	Hyperglycemia, infection, GI/mental effects; taper after prolonged therapy.
Montelukast	Maintenance/allergic asthma	Not rescue; monitor mood/behavior changes.
Omalizumab	Allergic asthma prevention	Anaphylaxis risk; administer and observe according to protocol.
Acetylcysteine	Mucolytic; acetaminophen antidote	Sulfur odor expected; bronchospasm can occur.

INHALER SEQUENCE Use the bronchodilator first when both are ordered, then the inhaled corticosteroid. Rinse the mouth after corticosteroid use.

TB therapy: RIPE

Drug	High-yield toxicity / teaching
Rifampin	Orange-red secretions, hepatotoxicity and many interactions; reduces hormonal contraceptive effectiveness.
Isoniazid	Hepatotoxicity and neuropathy; pyridoxine (B6) often prescribed.
Pyrazinamide	Hepatotoxicity and hyperuricemia.
Ethambutol	Optic neuritis; monitor acuity and color vision.

3. Anticoagulants, Antiplatelets and Thrombolytics

Drug/class	Monitoring / antidote	Safety
Unfractionated heparin	aPTT or anti-Xa; protamine	IV/SQ; monitor bleeding and HIT/platelets.
Enoxaparin	Usually no routine aPTT; protamine partially reverses	SQ abdomen; do not expel air bubble or massage. Renal dose adjustment.
Warfarin	PT/INR; vitamin K	Consistent vitamin K intake; many interactions; contraindicated in pregnancy.
DOACs	Drug-specific renal monitoring; idarucizumab for dabigatran, andexanet for factor Xa inhibitors in selected bleeding	Short half-life makes adherence crucial.
Aspirin	No routine level; supportive care/alkalinization for toxicity	Bleeding, GI injury, tinnitus; avoid in children with viral illness.
Clopidogrel	Clinical bleeding/CBC	Hold before procedures only as ordered.
Alteplase/tenecteplase	Frequent neuro/bleeding assessment	Strict eligibility and time windows; avoid invasive procedures after administration.

HIT A platelet fall after heparin may indicate heparin-induced thrombocytopenia. Stop all heparin and notify the provider; do not start warfarin alone in acute HIT.

4. Diuretics and Electrolytes

Class	Examples	Key adverse effects
Loop	Furosemide, bumetanide	Hypokalemia, dehydration, hypotension, ototoxicity; monitor weight, BP, renal function and electrolytes.
Thiazide	Hydrochlorothiazide, chlorthalidone	Hyponatremia, hypokalemia, hyperglycemia, hyperuricemia and hypercalcemia.
Potassium-sparing	Spironolactone, eplerenone, amiloride	Hyperkalemia; spironolactone may cause gynecomastia. Avoid potassium supplements/salt substitutes unless prescribed.
Osmotic	Mannitol	Pulmonary edema and fluid shifts; use filter and assess solution/crystals per product instructions.
Carbonic anhydrase inhibitor	Acetazolamide	Metabolic acidosis, hypokalemia and renal stones; used for glaucoma/altitude illness.

POTASSIUM Never give potassium IV push. Dilute and infuse with a pump according to policy; verify urine output and renal function.

5. Antihypertensives and Antianginals

Class	Examples	NCLEX focus
ACE inhibitor	Lisinopril, enalapril	Cough, hyperkalemia, angioedema, renal dysfunction; avoid pregnancy.
ARB	Losartan, valsartan	Hyperkalemia/angioedema; avoid pregnancy; less cough than ACE inhibitors.
Beta blocker	Metoprolol, propranolol	Bradycardia, hypotension; masks hypoglycemia. Nonselective agents may cause bronchospasm. Do not stop abruptly.
Calcium-channel blocker	Amlodipine, diltiazem, verapamil	Edema/hypotension; non-dihydropyridines slow HR/AV conduction and can cause constipation.
Alpha1 blocker	Prazosin, tamsulosin	First-dose/orthostatic hypotension; bedtime dosing may reduce falls.
Central alpha2 agonist	Clonidine, methyldopa	Sedation, dry mouth; abrupt clonidine withdrawal can cause rebound hypertension.
Direct vasodilator	Hydralazine, minoxidil	Reflex tachycardia/fluid retention; hydralazine may cause lupus-like syndrome.
Nitrate	Nitroglycerin	Headache/hypotension; never combine with PDE5 inhibitors. Store/administer correctly.

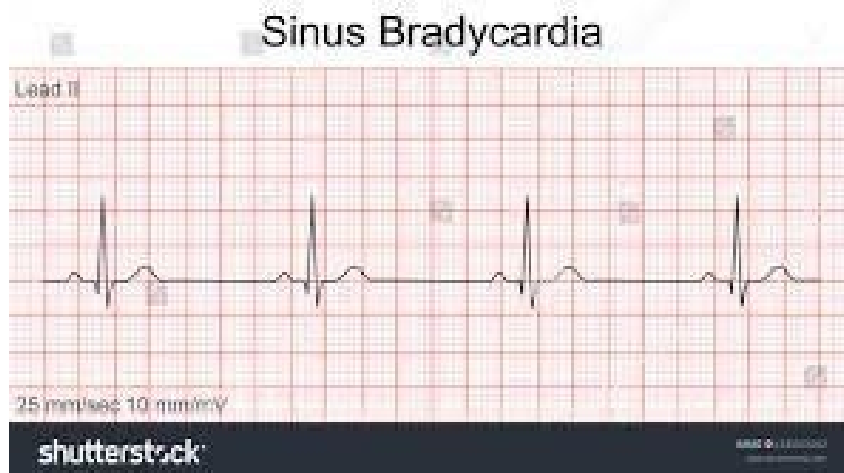
HYPERTENSIVE EMERGENCY Reduce BP with titratable IV medication and continuous monitoring; overly rapid reduction can cause cerebral, coronary or renal ischemia.

6. Heart Failure and Digoxin

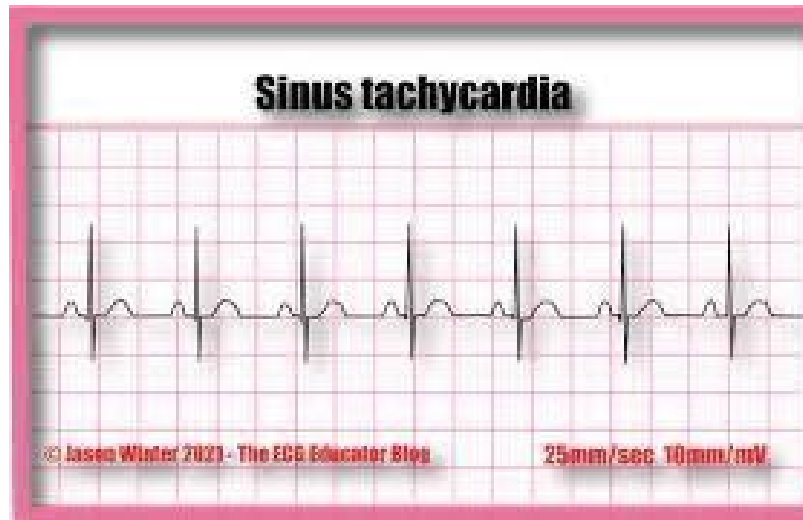
Medication	Role / monitoring
ACE inhibitor/ARB/ARNI	Reduces remodeling and events; monitor BP, potassium and renal function. Do not combine ACE inhibitor with sacubitril/valsartan without required washout.
Evidence-based beta blocker	Carvedilol, metoprolol succinate or bisoprolol for stable chronic HFrEF; begin low and monitor.
Loop diuretic	Relieves congestion; daily weights, I&O, electrolytes and renal function.
Aldosterone antagonist	Spirolactone/eplerenone; monitor potassium and renal function.
Digoxin	Improves symptoms/rate control but has narrow range. Check apical pulse, renal function, potassium and level when indicated.

DIGOXIN TOXICITY Anorexia, nausea/vomiting, confusion, visual changes and dysrhythmias are warning signs. Hypokalemia and renal impairment increase risk. Hold and notify; digoxin immune Fab treats severe toxicity.

7. Dysrhythmia Medications



Sinus bradycardia: rate is slow with a P wave before each QRS



Sinus tachycardia: rate is fast with a P wave before each QRS

Medication	Primary use / key risk
Adenosine	Stable regular narrow-complex SVT; rapid IV push followed by flush. Brief asystole/flushing/chest pressure expected; continuous ECG.
Atropine	Symptomatic bradycardia; may be ineffective in some high-grade blocks.
Amiodarone	Atrial/ventricular dysrhythmias; QT prolongation, bradycardia, pulmonary, thyroid, liver, eye and skin toxicity.
Lidocaine	Ventricular dysrhythmias; neurologic toxicity and seizures at high levels.
Diltiazem/verapamil	Rate control for AF/flutter and some SVT; bradycardia, block, hypotension.
Procainamide/sotalol	Selected arrhythmias; QT prolongation/torsades risk.

ALGORITHM Unstable tachycardia with a pulse generally requires synchronized cardioversion; ventricular fibrillation or pulseless VT requires defibrillation and CPR.

8. Anxiety, Sedation and Sleep

Class	Examples	Safety
Benzodiazepines	Lorazepam, diazepam, midazolam	Sedation, falls and respiratory depression, especially with opioids/alcohol. Dependence/withdrawal; taper. Flumazenil may provoke seizures.
Nonbenzodiazepine hypnotics	Zolpidem, eszopiclone	Complex sleep behaviors, next-day impairment and falls; take immediately before bed.
Buspirone	Chronic anxiety	Delayed onset; not for acute panic. Dizziness/nausea; serotonin interactions.
Ramelteon	Insomnia	Melatonin-receptor agonist; less dependence but still may impair alertness.

9. Mood Stabilizers and Antidepressants

Class/drug	Key teaching / toxicity
Lithium	Maintain consistent sodium/fluid intake. Monitor level, renal and thyroid function. Vomiting/diarrhea, coarse tremor, ataxia, confusion or seizures suggest toxicity. Avoid pregnancy and interacting NSAIDs/diuretics unless directed.
Valproate	Hepatotoxicity, pancreatitis, thrombocytopenia and teratogenicity; monitor liver function/CBC.
Carbamazepine	Agranulocytosis, hyponatremia, liver toxicity and serious rash; CBC/LFT/sodium, genetic screening when indicated.
SSRI/SNRI	Delayed benefit; nausea, insomnia, sexual dysfunction, bleeding and hyponatremia. Monitor early suicidality and serotonin syndrome.
TCA	Anticholinergic effects, orthostasis, sedation and lethal cardiac toxicity in overdose.
MAOI	Hypertensive crisis with tyramine/sympathomimetics and serotonin syndrome with serotonergic drugs; observe washout periods.
Bupropion	Less sexual dysfunction; contraindicated with seizure/eating disorders.
Trazodone	Sedation and orthostasis; priapism is an emergency.

SEROTONIN SYNDROME Agitation, hyperreflexia/clonus, diaphoresis, fever and diarrhea: stop serotonergic agents and provide emergency supportive care; cyproheptadine may be used.

10. Antipsychotics and Neuroleptic Malignant Syndrome

Group	Examples	Key adverse effects
First generation	Haloperidol, chlorpromazine	More EPS/tardive dyskinesia; anticholinergic effects, orthostasis and QT prolongation.
Second generation	Risperidone, olanzapine, quetiapine	Metabolic syndrome/weight gain, glucose/lipids; variable EPS/QT/prolactin effects.
Clozapine	Treatment-resistant schizophrenia	Severe neutropenia, myocarditis, seizures and metabolic effects; mandatory ANC monitoring.

EPS	Recognition / response
Acute dystonia	Painful neck/eye/jaw spasm; treat urgently with benztropine or diphenhydramine.
Akathisia	Inner restlessness; dose adjustment, beta blocker or other treatment.
Parkinsonism	Tremor, rigidity, bradykinesia; benztropine/amantadine may help.
Tardive dyskinesia	Late involuntary oral/limb movements; may be irreversible. Screen routinely; VMAT2 inhibitor may be used.

NMS EMERGENCY Severe rigidity, high fever, autonomic instability and altered consciousness: stop antipsychotic, transfer for emergency care, cool/hydrate and treat with dantrolene or bromocriptine as ordered.

11. Neurologic and Cognitive Medications

Medication	Use / high-yield teaching
Methylphenidate/amphetamines	ADHD; give early day, monitor BP/HR, appetite, weight/growth and misuse.
Donepezil/rivastigmine/galantamine	Mild-moderate Alzheimer symptoms; GI upset, bradycardia and syncope.
Memantine	Moderate-severe Alzheimer disease; dizziness/confusion; renal dose considerations.
Levodopa/carbidopa	Parkinson disease; nausea, orthostasis, dyskinesia and on-off fluctuations. Protein may reduce absorption in some patients.
Phenytoin	Seizures; nystagmus/ataxia toxicity, gingival hyperplasia, rash and interactions. IV only in compatible fluid and monitor cardiac status.
Baclofen/dantrolene	Spasticity; sedation/weakness. Dantrolene also treats malignant hyperthermia and may cause hepatotoxicity.

12. Maternity and Newborn Medications

Medication	Use	Priority
Magnesium sulfate	Seizure prevention/treatment in severe preeclampsia; sometimes fetal neuroprotection	Monitor RR, reflexes, urine output and level when ordered. Calcium gluconate for toxicity.
Oxytocin	Induction/augmentation and postpartum hemorrhage	Continuous fetal/uterine monitoring. Stop for tachysystole or nonreassuring fetal status; risk water intoxication.
Terbutaline	Short-term tocolysis in selected cases	Maternal tachycardia, hyperglycemia and pulmonary edema; not for prolonged maintenance.
Nifedipine	Tocolysis/hypertension	Hypotension and headache; monitor BP.
Methylergonovine	Postpartum uterine atony	Vasoconstriction; contraindicated in hypertension/preeclampsia.
Carboprost	Postpartum hemorrhage	Bronchospasm; avoid in asthma.
Betamethasone/dexamethasone	Accelerates fetal lung maturity	Give when preterm birth risk meets criteria; may increase glucose.
Rho(D) immune globulin	Prevents Rh sensitization	For eligible Rh-negative, unsensitized patient; verify blood type/antibody screen.
Erythromycin eye ointment / vitamin K	Newborn prophylaxis	Prevent gonococcal ophthalmia / hemorrhagic disease.

13. Musculoskeletal, Gout and Osteoporosis

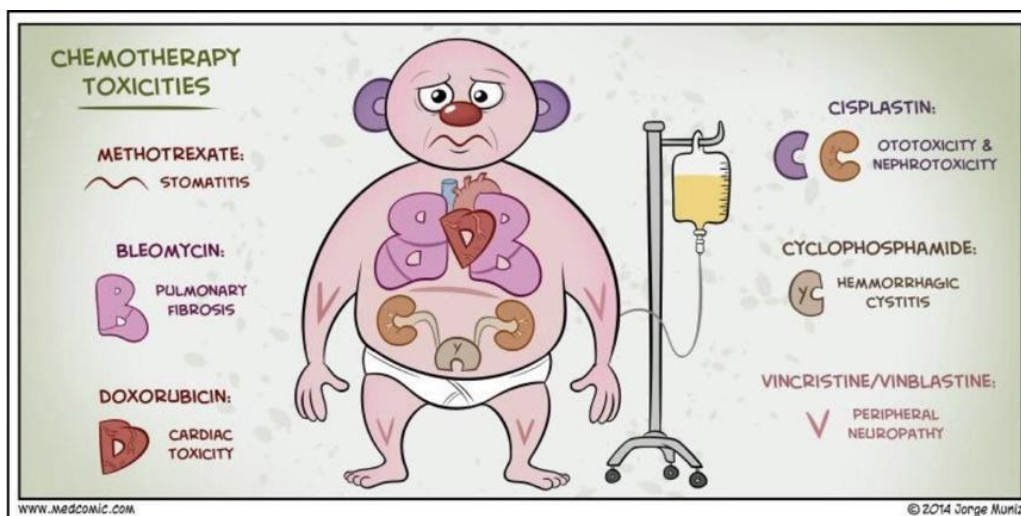
Medication	Key point
NSAIDs	GI bleeding, renal injury, fluid retention and cardiovascular risk; take as directed and avoid duplicate products.
Methotrexate	Rheumatoid arthritis/cancer; hepatotoxicity, marrow suppression, stomatitis and teratogenicity. Folic acid often used; avoid pregnancy/live vaccines.
Allopurinol	Prevents uric acid production; not immediate pain relief. Stop/report rash; interactions with azathioprine/mercaptopurine.
Colchicine	Acute gout inflammation; diarrhea is common and severe GI/myopathy/marrow toxicity can occur.
Bisphosphonates	Take first thing with plain water; remain upright at least 30-60 minutes; esophagitis and rare jaw osteonecrosis.
Calcitonin	Osteoporosis/hypercalcemia; nasal irritation and hypocalcemia.

14. Antibiotics and Antimicrobials

Class	Examples	High-yield adverse effects
Aminoglycosides	Gentamicin, amikacin, tobramycin	Nephrotoxicity and ototoxicity; levels/renal function.
Vancomycin	MRSA and serious gram-positive infection	Nephrotoxicity; infusion reaction if too rapid. Monitor exposure per protocol.
Fluoroquinolones	Ciprofloxacin, levofloxacin	Tendon rupture, neuropathy, CNS effects, QT prolongation and dysglycemia; separate from antacids/minerals.
Tetracyclines	Doxycycline	Photosensitivity, esophagitis, tooth/bone effects; avoid pregnancy/young children; separate from minerals.
Macrolides	Azithromycin, clarithromycin	GI upset, QT prolongation, liver injury and interactions.
Penicillins/cephalosporins	Amoxicillin, ceftriaxone	Allergy/anaphylaxis, diarrhea/C. difficile; assess reaction history.
Sulfonamides	TMP-SMX	Rash/SJS, hyperkalemia, marrow suppression, crystalluria; photosensitivity and interactions.
Metronidazole	Anaerobes/protozoa	Metallic taste, neuropathy; avoid alcohol during and after therapy per product guidance.

CULTURES Obtain cultures before the first antibiotic dose when feasible, but do not delay urgent treatment in sepsis. Complete the prescribed course and report severe diarrhea.

15. Cancer Medications and Toxicity Map



Original chemotherapy-toxicity memory map

Agent/class	Signature toxicity / protection
Doxorubicin	Cardiomyopathy and vesicant injury; monitor cardiac function and lifetime dose. Dexrazoxane may protect.
Cisplatin	Nephrotoxicity and ototoxicity; hydration, renal function and hearing.
Cyclophosphamide	Hemorrhagic cystitis; hydration and mesna.
Methotrexate	Marrow/liver/mucosal toxicity; leucovorin rescue for high-dose therapy.
Bleomycin	Pulmonary fibrosis; monitor respiratory symptoms and cumulative exposure.
Vincristine	Peripheral/autonomic neuropathy and ileus; fatal if given intrathecally.
Paclitaxel	Hypersensitivity and neuropathy; premedication as ordered.
Tamoxifen	VTE and endometrial cancer risk; report leg swelling or abnormal bleeding.

EXTRAVASATION Stop the infusion, leave the catheter in place to aspirate/administer antidote if ordered, notify the oncology team, and apply the drug-specific compress. Do not flush.

16. Hematopoietic Growth Factors and Biologics

Drug	Use / monitoring
Filgrastim/pegfilgrastim	Raises neutrophils; bone pain and rare splenic rupture. Avoid administration too close to cytotoxic chemotherapy per protocol.
Epoetin alfa/darbepoetin	Stimulates RBCs in CKD/selected chemotherapy anemia; hypertension and thrombosis. Use lowest effective dose and ensure iron.
Oprelvekin	Raises platelets in selected settings; fluid retention and dysrhythmias.
Interferons	Antiviral/antitumor effects; flu-like symptoms, depression, marrow and liver toxicity.
Monoclonal antibodies	Target-specific; infusion reactions, infection/reactivation and organ-specific toxicity. Screen/monitor by product.

NADIR The lowest blood-cell count after chemotherapy is the nadir. During neutropenia, fever is an emergency; avoid infection exposures and follow neutropenic precautions.

17. Rapid Review: Interactions and Red Flags

Finding	Most likely medication concern
Tinnitus after aspirin	Salicylate toxicity.
Heel/tendon pain	Fluoroquinolone-associated tendon injury.
Sore throat/fever on clozapine	Neutropenia/agranulocytosis.
Dry cough or swollen lips/tongue	ACE inhibitor cough or angioedema.
Orange urine/tears	Expected rifampin effect.
Coarse tremor, ataxia, diarrhea	Lithium toxicity.
Rigidity + fever on antipsychotic	Neuroleptic malignant syndrome.
Clonus + fever on serotonergic drugs	Serotonin syndrome.
Petechiae/platelet fall on heparin	HIT.
Visual halos, nausea, dysrhythmia	Digoxin toxicity.
Absent reflexes, slow respirations on magnesium	Magnesium toxicity.
Pain/burning at vesicant IV site	Chemotherapy extravasation.

FINAL NCLEX FOCUS Prioritize airway, breathing, circulation, bleeding, severe allergy, neurologic change and organ toxicity. Verify the medication against current orders and facility policy; never rely on a mnemonic alone for administration decisions.